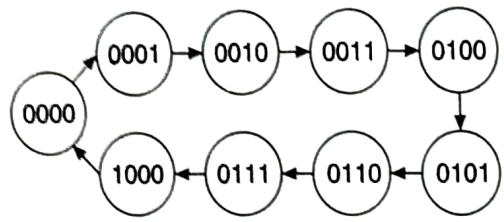


12.5.6 Design a Modulo-9 Synchronous Counter Using T FFs

Step 1. The number of flip-flops: We know that the counting sequence for a modulo-9 counter is 0000, 0001, 0010, 0011, 0100, 0101, 0110, 0111, 1000, 0000, It has 9 states. So it requires 4 FFs($9 \leq 2^4$). 4 flip-flops can have 16 states. 7 states 1001, 1010, 1011, 1100, 1101, 1110, and 1111 are invalid. The entries for excitations corresponding to invalid states are don't cares.

Step 2. The state diagram: The state diagram for the mod-9 counter is drawn as shown in Figure 12.23a.

Step 3. The type of flip-flops and the excitation table: T flip-flops are selected and the excitation table of a mod-9 counter using T FFs is drawn as shown in Figure 12.23b.



(a) State diagram

PS				NS				Required Excitations			
Q ₄	Q ₃	Q ₂	Q ₁	Q ₄	Q ₃	Q ₂	Q ₁	T ₄	T ₃	T ₂	T ₁
0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	1	0	0	1	0	0	0	1	1
0	0	1	0	0	0	1	1	0	0	0	1
0	0	1	1	0	1	0	0	0	1	1	1
0	1	0	0	0	1	0	1	0	0	0	1
0	1	0	1	0	1	1	0	0	0	1	1
0	1	1	0	0	1	1	1	0	0	0	1
0	1	1	1	1	0	0	0	1	1	1	1
1	0	0	0	0	0	0	0	1	0	0	0

(b) Excitation table

Figure 12.23 A synchronous mod-9 counter.

Step 4. The minimal expressions: The K-maps for excitations T_4 , T_3 , T_2 , and T_1 in terms of the outputs of the FFs Q_4 , Q_3 , Q_2 , and Q_1 , their minimization and the minimal expressions for excitations obtained from them are shown in Figure 12.24.

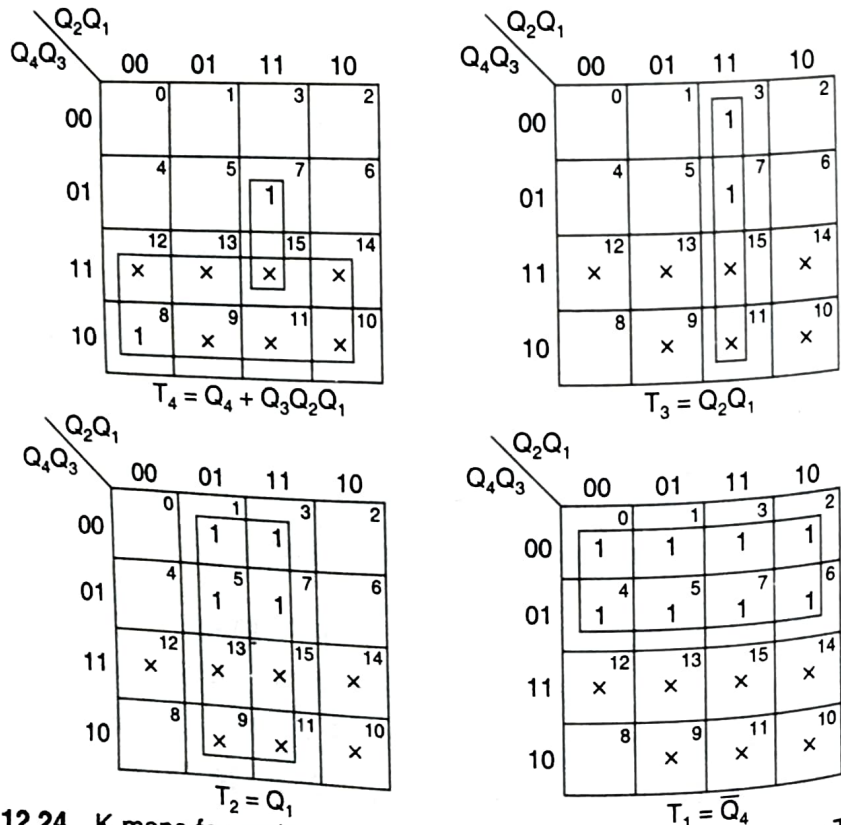


Figure 12.24 K-maps for excitations of synchronous mod-9 counter using T flip-flops.

Step 5. The logic diagram: The logic diagram based on those minimal expressions is drawn as shown in Figure 12.25.

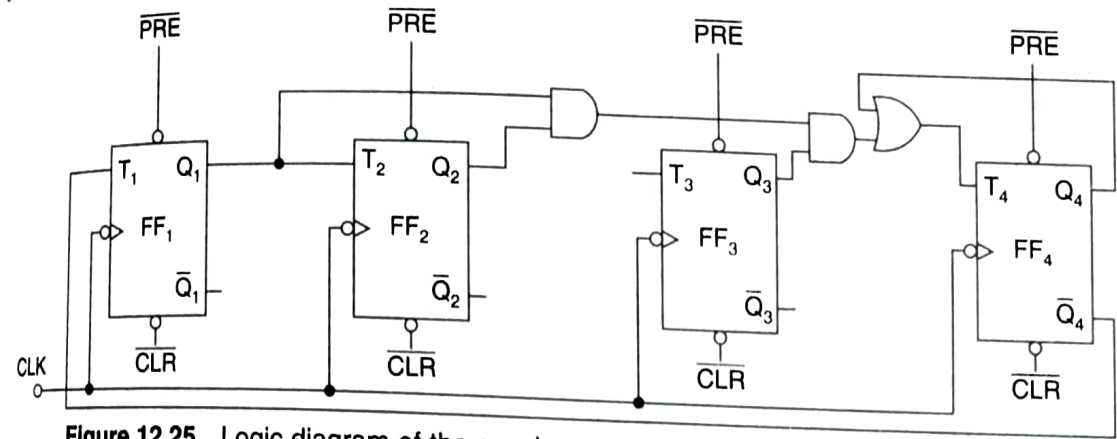


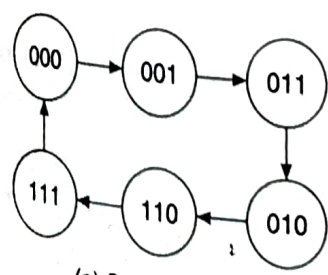
Figure 12.25 Logic diagram of the synchronous mod-9 counter using T flip-flops.

12.5.7 Design of a Synchronous Modulo-6 Gray Code Counter

Step 1. The number of flip-flops: We know that the counting sequence for a modulo-6 Gray code counter is 000, 001, 011, 010, 110 and 111. It requires $n = 3$ FFs ($N \leq 2^n$, i.e. $6 \leq 2^3$). 3 FFs can have 8 states. So the remaining two states 101 and 100 are invalid. The entries for excitations corresponding to invalid states are don't cares.

Step 2. The state diagram: The state diagram of the mod-6 Gray code counter is drawn as shown in Figure 12.26a.

Step 3. Type of flip-flops and the excitation table: T flip-flops are selected and the excitation table of the mod-6 Gray code counter using T flip-flops is written as shown in Figure 12.26b.



PS			NS			Required Excitations		
Q ₃	Q ₂	Q ₁	Q ₃	Q ₂	Q ₁	T ₃	T ₂	T ₁
0	0	0	0	0	1	0	0	1
0	0	1	0	1	1	0	1	0
0	1	1	0	1	0	0	0	1
0	1	0	1	1	0	1	0	0
1	1	0	1	1	1	0	0	1
1	1	1	0	0	0	1	1	1

(a) State diagram

(b) Excitation table

Figure 12.26 Synchronous mod-6 Gray code counter.

Step 4. The minimal expressions: The K-maps for excitations of FFs T₃, T₂ and T₁ in terms of outputs of FFs Q₃, Q₂ and Q₁, their minimization and the minimal expressions for excitations obtained from them are shown in Figure 12.27.

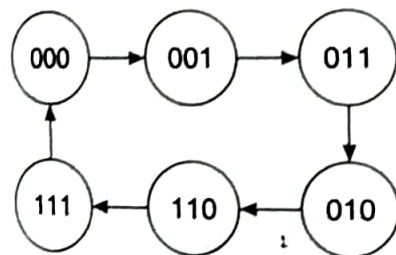
Step 5. The logic diagram: The logic diagram based on those minimal expressions is drawn as shown in Figure 12.28.

12.5.7 Design of a Synchronous Modulo-6 Gray Code Counter

Step 1. The number of flip-flops: We know that the counting sequence for a modulo-6 Gray code counter is 000, 001, 011, 010, 110 and 111. It requires $n = 3$ FFs ($N \leq 2^n$, i.e. $6 \leq 2^3$). 3 FFs can have 8 states. So the remaining two states 101 and 100 are invalid. The entries for excitations corresponding to invalid states are don't cares.

Step 2. The state diagram: The state diagram of the mod-6 Gray code counter is drawn as shown in Figure 12.26a.

Step 3. Type of flip-flops and the excitation table: T flip-flops are selected and the excitation table of the mod-6 Gray code counter using T flip-flops is written as shown in Figure 12.26b.



(a) State diagram

PS			NS			Required Excitations		
Q_3	Q_2	Q_1	Q_3	Q_2	Q_1	T_3	T_2	T_1
0	0	0	0	0	1	0	0	1
0	0	1	0	1	1	0	1	0
0	1	1	0	1	0	0	0	1
0	1	0	1	1	0	1	0	0
1	1	0	1	1	1	0	0	1
1	1	1	0	0	0	1	1	1

(b) Excitation table

Figure 12.26 Synchronous mod-6 Gray code counter.

Step 4. The minimal expressions: The K-maps for excitations of FFs T_3 , T_2 and T_1 in terms of outputs of FFs Q_3 , Q_2 and Q_1 , their minimization and the minimal expressions for excitations obtained from them are shown in Figure 12.27.

Step 5. The logic diagram: The logic diagram based on those minimal expressions is drawn as shown in Figure 12.28.

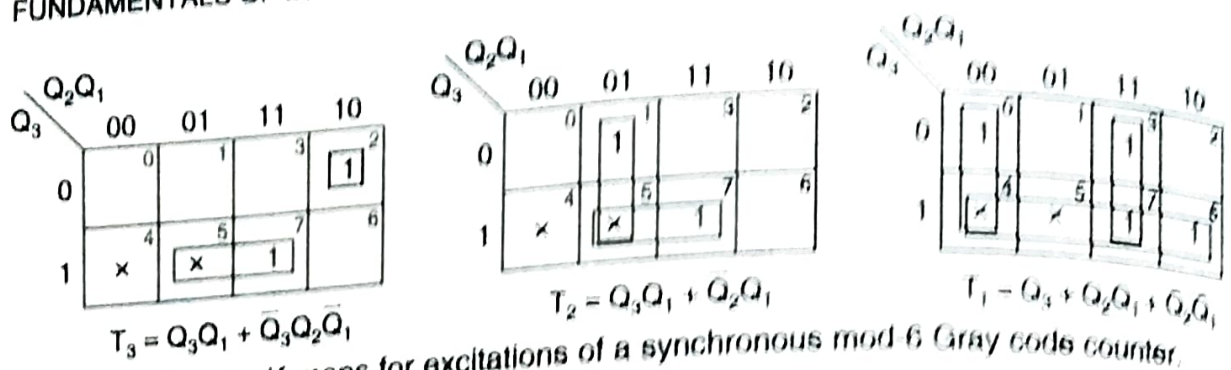


Figure 12.27 K-maps for excitations of a synchronous mod 6 Gray code counter.

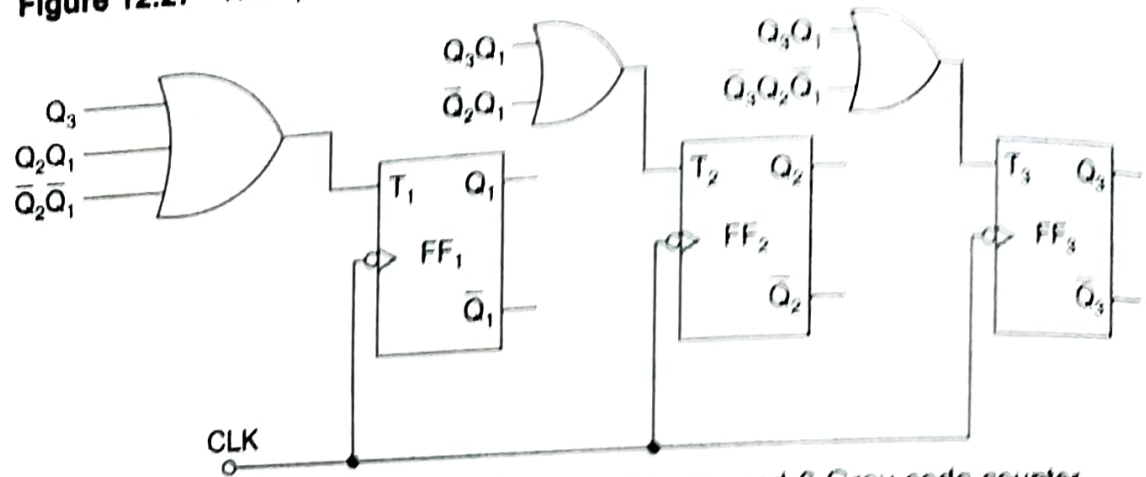


Figure 12.28 Logic diagram of a synchronous mod-6 Gray code counter.

12.5.8 Design of a Synchronous Modulo-10 Gray Code Counter

Step 1. The number of flip-flops: We know that the counting sequence for a modulo-10 Gray code counter is 0000, 0001, 0011, 0010, 0110, 0111, 0101, 0100, 1100, 1101, 0000.... It has 10 states and so it requires $n = 4$ FFs ($N \leq 2^n$, i.e. $10 \leq 2^4$). Four FFs can have 16 states. So the remaining six states 1111, 1110, 1010, 1011, 1001 and 1000 are invalid. The entries for excitations corresponding to these invalid states are don't cares.

Step 2. The state diagram: The state diagram of the mod-10 Gray code counter is drawn as shown in Figure 12.29a.

Step 3. The type of flip-flops and the excitation table: T flip-flops are selected and the excitation table of the mod-10 Gray code counter using T FFs is written as shown in Figure 12.29b.

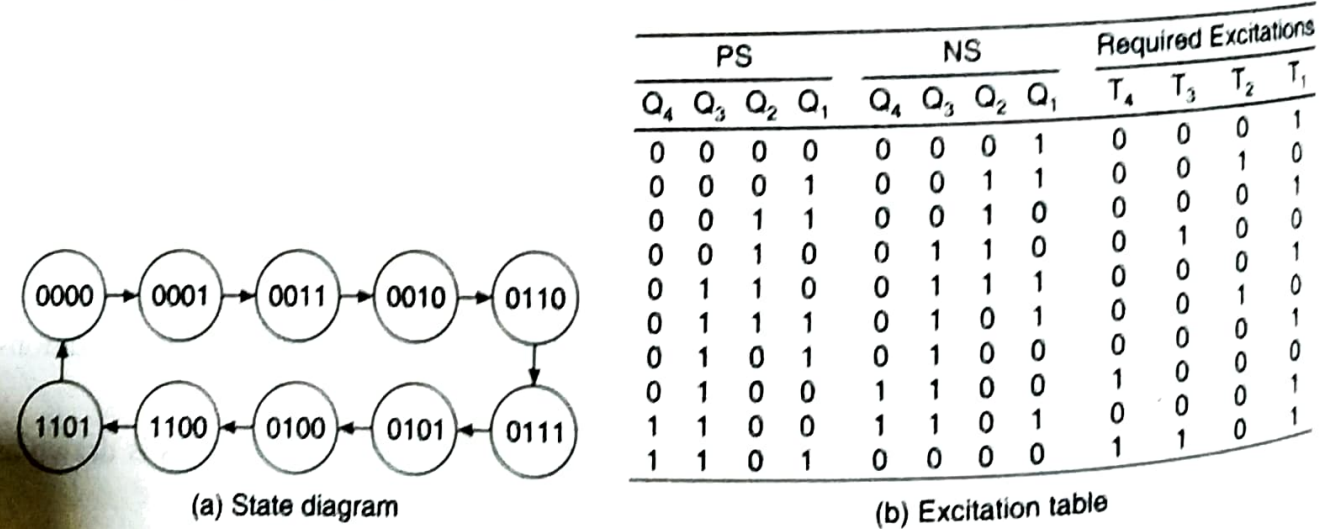


Figure 12.29 Synchronous mod-10 Gray code counter.

Step 4. The minimal expressions: The K-maps for excitations of FFs T_4 , T_3 , T_2 and T_1 in terms of outputs of FFs Q_4 , Q_3 , Q_2 and Q_1 , their minimization, and the minimal expressions for excitations obtained from them are shown in Figure 12.30.

Step 5. The logic diagram: The logic diagram based on those minimal expressions is drawn as shown in Figure 12.31.

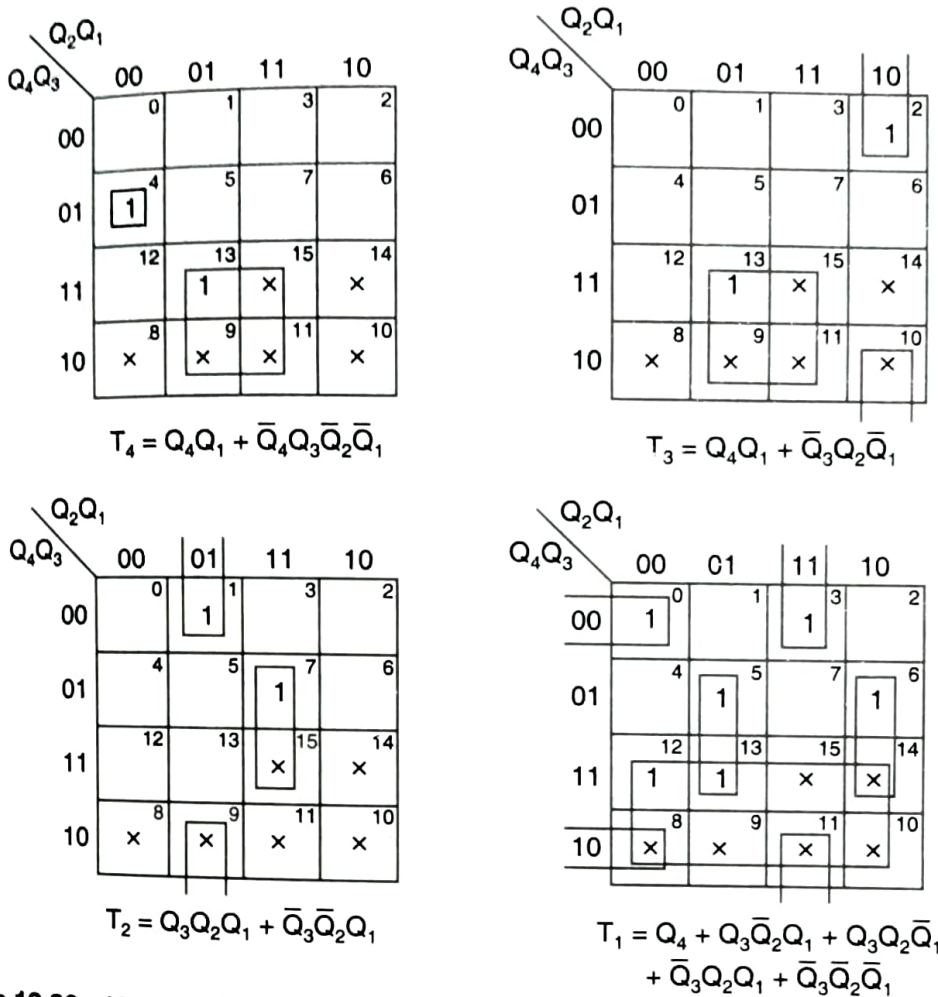


Figure 12.30 K-maps for excitations of a synchronous mod-10 Gray code counter.

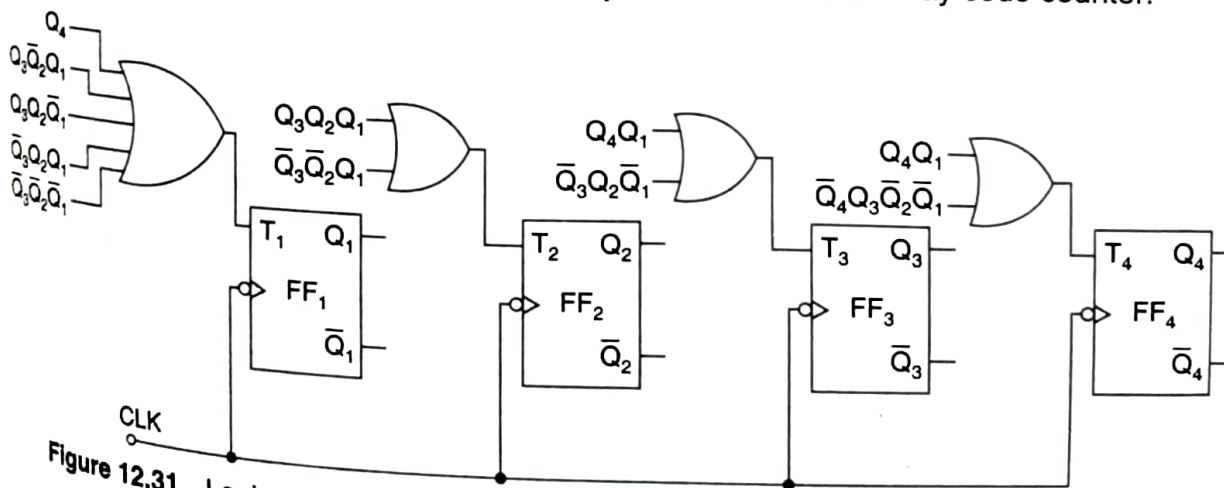


Figure 12.31 Logic diagram of synchronous mod-10 Gray code counter using T FFs.

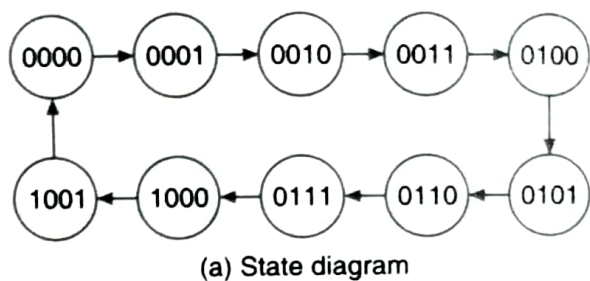
12.5.9 Design of a Synchronous BCD Counter Using J-K FFs

Step 1. The number of flip-flops: A BCD counter is nothing but a mod-10 counter. It is a decade counter. It has 10 states (0000 through 1001). It requires $n = 4$ FFs ($N \leq 2^n$, i.e. $10 \leq 2^4$). Four FFs can have 16 states. After the tenth clock pulse, the counter resets. So, states 1010 through 1111 are invalid. The entries for excitations corresponding to invalid states are don't cares.

Step 2. The state diagram: The state diagram of the BCD counter is drawn as shown in Figure 12.32a.

Step 3. The type of flip-flops and the excitation table: JK flip-flops are selected and the excitation table of the BCD counter using J-K FFs is drawn as shown in Figure 12.32b.

Step 4. The minimal expressions: The K-maps for the excitations of FFs $J_4, K_4, J_3, K_3, J_2, K_2, J_1$ and K_1 in terms of the outputs of the FFs Q_4, Q_3, Q_2 and Q_1 , based on the excitation table, their minimization and the minimal expressions obtained from them are shown in Figure 12.33.



PS				NS				Required excitations							
Q_4	Q_3	Q_2	Q_1	Q_4	Q_3	Q_2	Q_1	J_4	K_4	J_3	K_3	J_2	K_2	J_1	K_1
0	0	0	0	0	0	0	1	0	x	0	x	0	x	1	x
0	0	0	1	0	0	1	0	0	x	0	x	1	x	x	1
0	0	1	0	0	0	1	1	0	x	0	x	x	0	1	x
0	0	1	1	0	1	0	0	0	x	1	x	x	1	x	1
0	1	0	0	0	1	0	1	0	x	x	0	0	x	1	x
0	1	0	1	0	1	1	0	0	x	x	0	1	x	x	1
0	1	1	0	0	1	1	1	0	x	x	0	x	0	1	x
0	1	1	1	1	0	0	0	1	x	x	1	x	1	x	1
1	0	0	0	1	0	0	1	x	0	0	x	0	x	1	x
1	0	0	1	0	0	0	0	x	1	0	x	0	x	x	1

(b) Excitation table
Figure 12.32 Synchronous BCD (mod-10) counter.

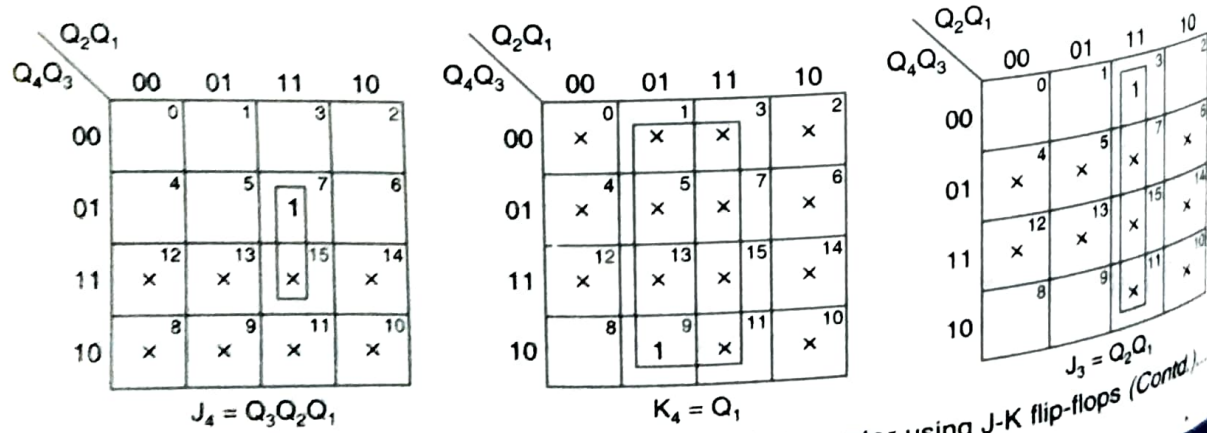


Figure 12.33 K-maps for excitations of synchronous BCD counter using J-K flip-flops (Contd.)

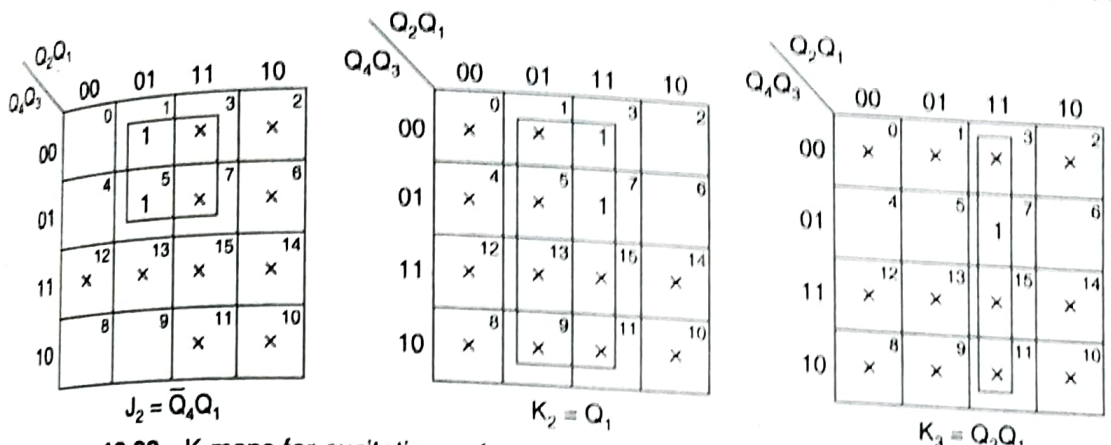


Figure 12.33 K-maps for excitations of synchronous BCD counter using J-K flip-flops.

Step 5. The logic diagram: The logic diagram based on those minimal expressions is drawn as shown in Figure 12.30.

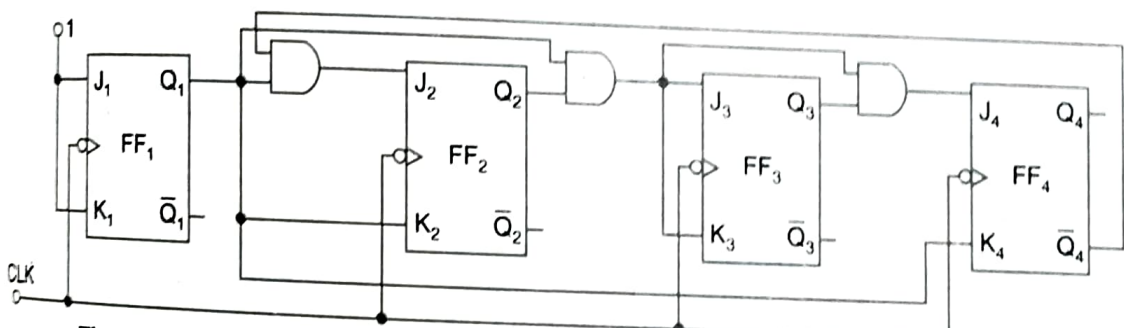
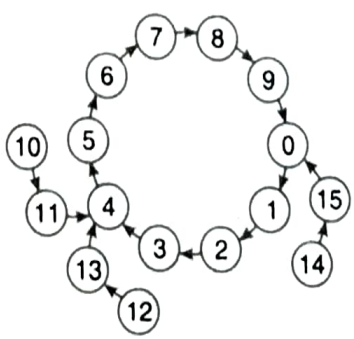


Figure 12.34 Logic diagram of synchronous BCD counter using J-K flip-flops.

The state diagram and the table to check for lock-out are shown in Figure 12.35. The table to check for lock-out shows that, if the counter finds itself in an invalid state initially, it moves to a valid state after one or two clock pulses and then counts in the normal way. Therefore, the counter is self-starting.

PS				Present inputs								NS			
Q_4	Q_3	Q_2	Q_1	J_4	K_4	J_3	K_3	J_2	K_2	J_1	K_1	Q_4	Q_3	Q_2	Q_1
1	0	1	0	0	0	0	0	0	0	1	1	1	0	1	1
1	0	1	1	0	1	1	1	0	1	1	1	0	1	0	0
1	1	0	0	0	0	0	0	0	0	1	1	1	1	0	1
1	1	0	1	0	1	0	0	0	1	1	1	0	1	0	0
1	1	1	0	0	0	0	0	0	0	1	1	1	1	1	1
1	1	1	1	1	1	1	1	0	1	1	1	0	0	0	0



(a) Table to check for lock-out

(b) State diagram

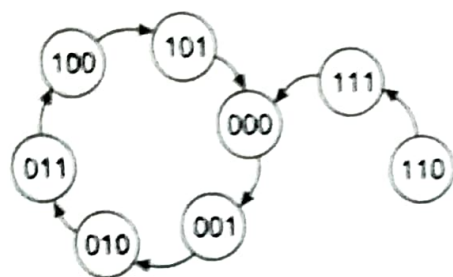
Figure 12.35 Checking for lock out of Synchronous BCD counter using J-K flip-flops.

12.5.10 Design of a Synchronous Mod-6 Counter Using J-K FFs

Step 1. The number of flip-flops: We know that the counting sequence for a mod-6 counter is 000, 001, 010, 011, 100, 101, 000, ... It has six states. So it requires $n = 3$ FFs ($N \leq 2^n$, i.e. $6 \leq 2^3$). Three FFs can have eight states. So the remaining two states 110 and 111 are invalid. The entries for excitations corresponding to invalid states are don't cares.

Step 2. The state diagram: The state diagram for the mod-6 counter is drawn as shown in Figure 12.36a.

Step 3. The type of flip-flops and the excitation table: JK flip-flops are selected and the excitation table of a mod-6 counter using J-K FFs is drawn as shown in Figure 12.36b. (States 110 and 111 can be removed from the state diagram if it is not required to determine whether the counter is self starting or not).



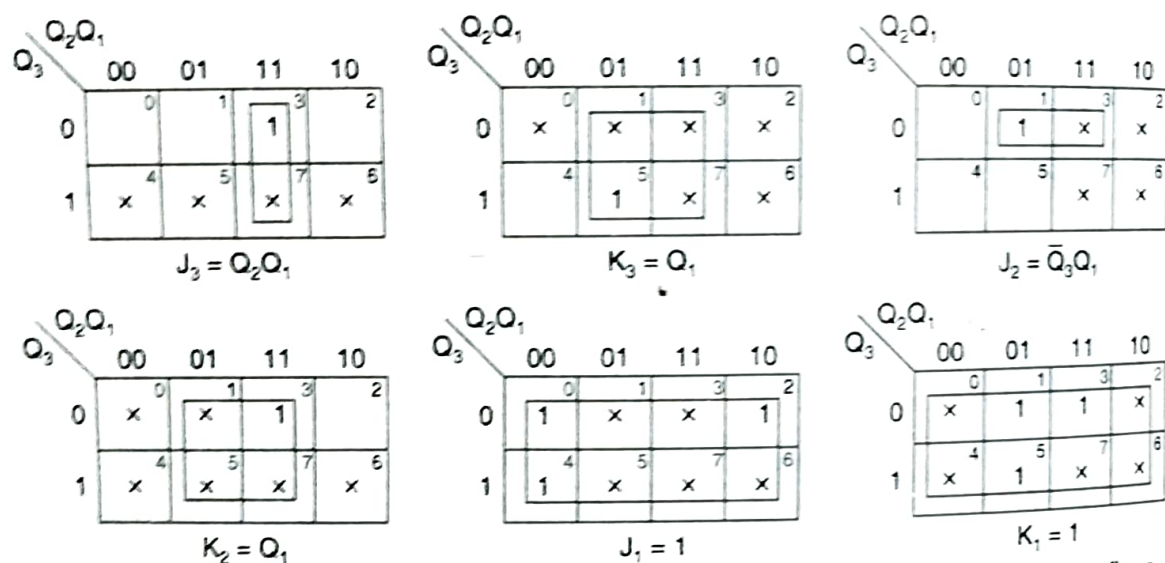
(a) State diagram

PS			NS			Required excitations				
Q_3	Q_2	Q_1	Q_3	Q_2	Q_1	J_3	K_3	J_2	K_2	J_1 , K_1
0	0	0	0	0	1	0	x	0	x	1 x
0	0	1	0	1	0	0	x	1	x	x 1
0	1	0	0	1	1	0	x	x	0	1 x
0	1	1	1	0	0	1	x	x	1	x 1
1	0	0	1	0	1	x	0	0	x	1 x
1	0	1	0	0	0	x	1	0	x	x 1

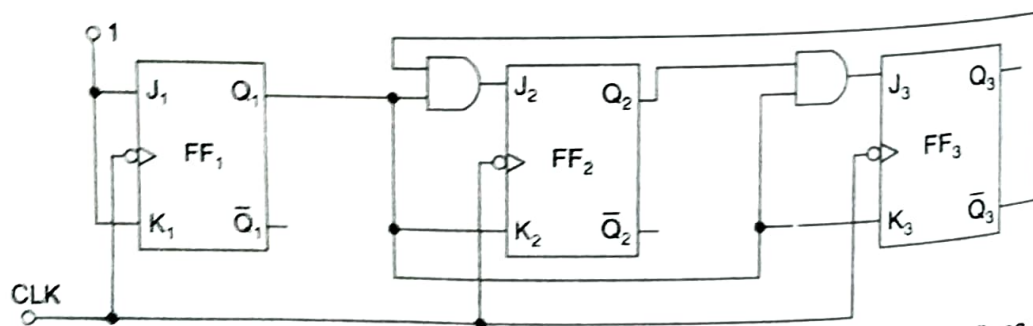
(b) Excitation table

Figure 12.36 Synchronous mod-6 counter using J-K flip-flops.

Step 4. The minimal expressions: The K-maps for excitations of FFs J_3 , K_3 , J_2 , K_2 , J_1 and K_1 in terms of the outputs of FFs Q_3 , Q_2 and Q_1 , their minimization, and the minimal expressions for excitations obtained from them are shown in Figure 12.37.

**Figure 12.37** K-maps for excitations of synchronous mod-6 counter using J-K flip-flops.

Step 5. The logic diagram: The logic diagram based on those minimal expressions is drawn as shown in Figure 12.38.

**Figure 12.38** Logic diagram of synchronous mod-6 counter using J-K flip-flops.

the counter. With these inputs applied to the counter, we can determine each of the next states of the counter. From Table 12.3 which is in the present state, we can determine the next states of the counter. From Table 12.3 we see that, if the present state ($Q_3Q_2Q_1$) is 110, the excitations are $J_3 = 0$, $K_3 = 0$, $J_2 = 0$, $K_2 = 0$, $J_1 = 1$ and $K_1 = 1$. With these excitations, the counter changes from 110 to 111. If the present state is 111, the excitations are $J_3 = 1$, $K_3 = 1$, $J_2 = 0$, $K_2 = 1$, $J_1 = 1$ and $K_1 = 1$. With these excitations the counter changes from 111 to 000. From the table, to check for lock-out, we see that the counter is self-starting because if the counter initially finds itself in state 111, it goes to 000 after one clock pulse and if the counter is initially in state 110, then it goes to 111 after one clock pulse and goes to 000 after two clock pulses.

Table 12.3 Check for lock-out

PS			Present inputs						NS		
Q_3	Q_2	Q_1	J_3	K_3	J_2	K_2	J_1	K_1	Q_3	Q_2	Q_1
1	1	0	0	0	0	0	1	1	1	1	1
1	1	1	1	1	0	1	1	1	0	0	0

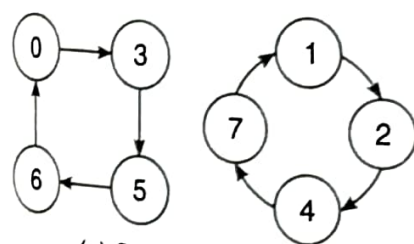
EXAMPLE 12.4 Design a type T counter that goes through states 0, 3, 5, 6, 0 Is the counter self-starting?

Solution

Step 1. The number of flip-flops: This counter has only four stable states, but it requires three FFs, because it counts 101 and 110 ($6 \leq 2^3$). Three FFs can have 8 states, out of which states 000, 011, 101, 110 are valid and states 001, 010, 100, 111 are invalid. The entries for excitations corresponding to invalid states are don't cares.

Step 2. The state diagram: The state diagram of the counter is shown in Figure 12.39a.

Step 3. The type of flip-flops and the excitation table: T flip-flops are selected and the excitation table of this counter using T flip-flops is drawn as shown in Figure 12.39b.



(a) State diagram

PS			NS			Required excitations		
Q_3	Q_2	Q_1	Q_3	Q_2	Q_1	T_3	T_2	T_1
0	0	0	0	1	1	0	1	1
0	1	1	1	0	1	1	1	0
1	0	1	1	1	0	0	1	1
1	1	0	0	0	0	1	1	0

(b) Excitation table

Figure 12.39 Example 12.4: T type, 0, 3, 5, 6, 0 ... counter.

Step 4. The minimal expressions: The K-maps for T_3 , T_2 , and T_1 in terms of Q_3 , Q_2 , and Q_1 based on the excitation table, their minimization, and the minimal expressions obtained from them are shown in Figure 12.40.

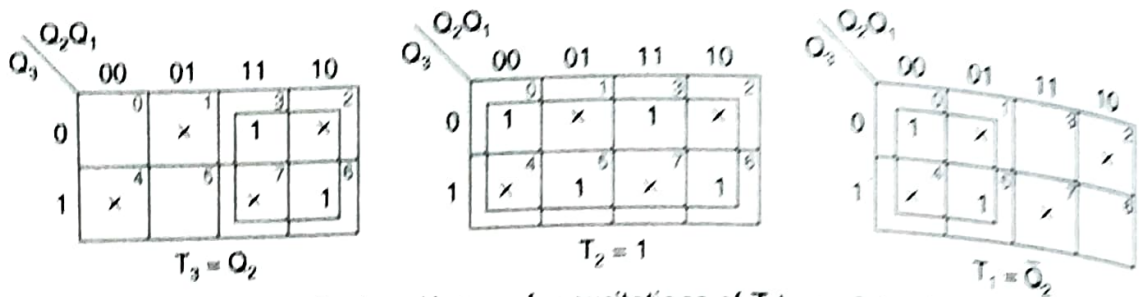
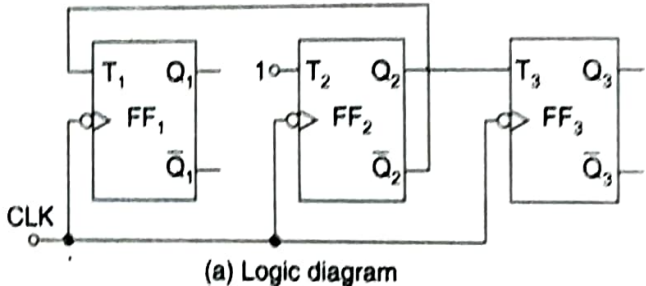


Figure 12.40 Example 12.4: K-maps for excitations of T type, 0, 3, 5, 6, 0... counter.

Step 5. The logic diagram: The logic diagram based on those minimal expressions is shown in Figure 12.41a.

The table to check for lock-out is shown in Figure 12.41b. We see that the counter is not self-starting. It suffers from the problem of lock-out. That is, initially if it enters an invalid state, it keeps on moving from one invalid state to another when clock pulses are applied and never returns to a valid state and, therefore, it serves no useful purpose.



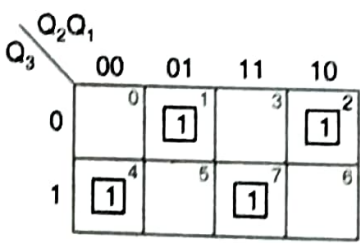
PS			Present inputs			NS		
Q ₃	Q ₂	Q ₁	T ₃	T ₂	T ₁	Q ₃	Q ₂	Q ₁
0	0	1	0	1	1	0	1	0
0	1	0	1	1	0	1	0	0
1	0	0	0	1	1	1	1	1
1	1	1	1	1	0	0	0	1

Figure 12.41 Example 12.4: T type counter that goes through states 0, 3, 5, 6, 0, ...

Elimination of lock-out: There are two methods to eliminate lock-out.

First method. Obtain a reset pulse as shown in Figure 12.42 such that, whenever the counter goes to an invalid state, it automatically resets to the initial state 000.

PS			Reset
Q ₃	Q ₂	Q ₁	R
0	0	1	1
0	1	0	1
1	0	0	1
1	1	1	1



$$R = \overline{Q}_3\overline{Q}_2Q_1 + \overline{Q}_3Q_2\overline{Q}_1 + Q_3\overline{Q}_2\overline{Q}_1 + Q_3Q_2Q_1$$
$$= \overline{Q}_3(Q_2 \oplus Q_1) + Q_3(\overline{Q}_2 \oplus \overline{Q}_1)$$
$$= Q_3 \oplus Q_2 \oplus Q_1$$

Figure 12.42 Example 12.4: Table for R, K-map, and the minimal expression for R.

Obtain this reset pulse from the counter already designed (taking the excitations corresponding to the invalid states as don't cares) and feed it to the CLR terminals of the FFs. The circuit of Figure 12.41a is, therefore, modified as shown in Figure 12.43. The counter comes out of the invalid state without requiring another clock pulse.

Second method. Redesign the counter assuming that the counter goes to the starting state, i.e. 000 whenever it enters any of the invalid states. The excitation requirements are shown in Figure 12.44a. In this method, no don't cares are available for excitations even if the present state is an invalid one. The counter designed by this method cannot come out of the invalid state

instantaneously. It requires one more clock pulse to let the counter come out of the invalid state. The circuit is also more complicated because no don't cares are present. The K-maps for T_3 , T_2 , and T_1 based on the excitation table and the minimal expressions obtained from them are shown in Figure 12.44b. A logic circuit may be realized using those minimal expressions.

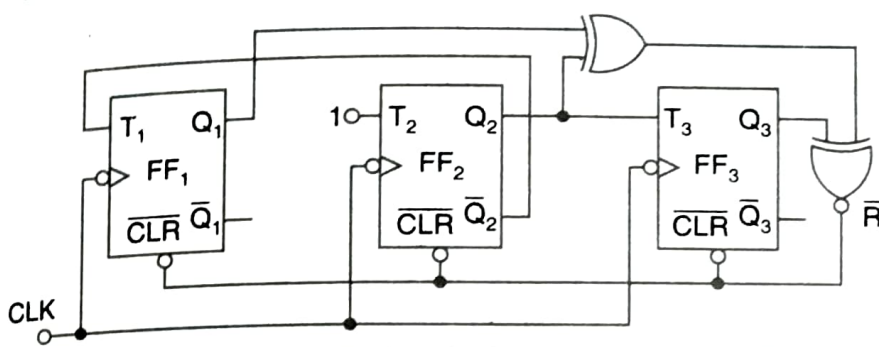


Figure 12.43 Logic diagram of the type T 0, 3, 5, 6, 0... counter modified to eliminate lock-out.

PS			NS			Required excitations		
Q_3	Q_2	Q_1	Q_3	Q_2	Q_1	T_3	T_2	T_1
0	0	0	0	1	1	0	1	1
0	0	1	0	0	0	0	0	1
0	1	0	0	0	0	0	1	0
0	1	1	1	0	1	1	1	0
1	0	0	0	0	0	1	0	0
1	0	1	1	1	0	0	1	1
1	1	0	0	0	0	1	1	0
1	1	1	0	0	0	1	1	1

(a) Excitation table

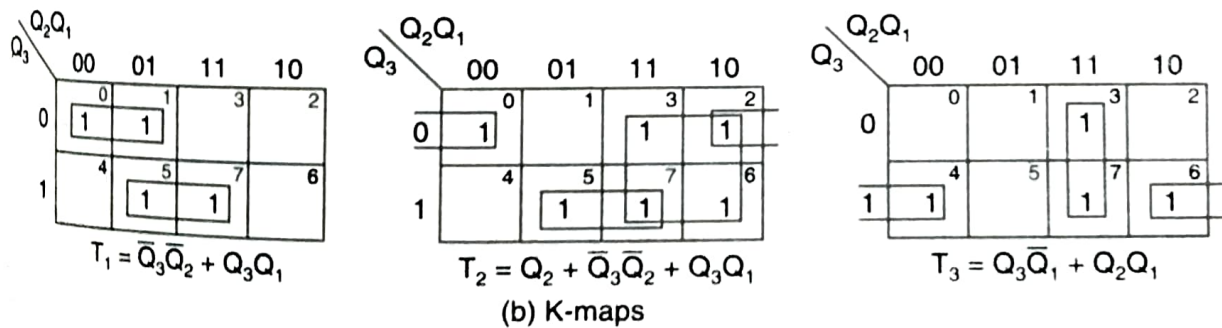


Figure 12.44 Example 12.4: Excitation table and K-maps.

EXAMPLE 12.5 Design a type D counter that goes through states 0, 1, 2, 4, 0, The undesired (unused) states must always go to zero (000) on the next clock pulse.

Solution

Solution

Step 1. The number of flip-flops: This counter has only four stable states 0, 1, 2 and 4 (000, 001, 010, 100), but it requires three FFs because it counts 4 (100) as well. Three FFs can have eight states. So, the remaining four states (011, 101, 110, 111) are undesired. These undesired states must go to 000 after the next clock pulse. So, no don't cares.

Step 2. The state diagram: The state diagram of the 0, 1, 2, 4, 0, ... counter is drawn as shown in Figure 12.45a.